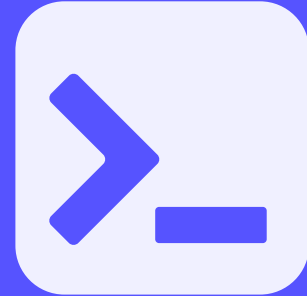






# PYTHON CHEATSHEET FOR ABSOLUTE BEGINNERS



## Introduction to Functions

### Function

A function is a reusable block of code that performs a specific task or a set of related tasks.

### Syntax of a Function

```
def function_name(parameters):  
    # Function body  
    # ...  
    return result
```

- **def**: Keyword to define a function
- **function\_name**: Name of the function
- **parameters**: Input values to the function
- **return**: Keyword to return a result (optional)

**Note:** A function body comprises a set of instructions that compute the result based on the input parameters.

### Function Calls

```
result = function_name(arguments)
```

- **function\_name**: Name of the function
- **arguments**: Values passed as function parameters
- **result**: Result returned by the function (if applicable)

### Practice: Creating and Calling Functions

```
def greet(name):  
    message = f"Hello, {name}!"  
    return message  
  
message = greet("Alice")  
print(message) # Output: Hello, Alice!
```

## Python's Built-in Toolbox

### Built-in Functions for Strings

#### The `is...` methods

- `isalpha()`: Checks if all characters are alphabetic
- `isdigit()`: Checks if all characters are digits
- `isalnum()`: Checks if all characters are alphanumeric
- `islower()`: Checks if all characters are lowercase
- `isupper()`: Checks if all characters are uppercase

#### Methods related to case change

- `lower()`: Converts all characters to lowercase
- `upper()`: Converts all characters to uppercase
- `title()`: Converts the first character of each word to uppercase

#### Search-related methods

- `find()`: Returns the index of the first occurrence
- `count()`: Returns the number of occurrences
- `startswith()`: Checks if the string starts with a specified prefix
- `endswith()`: Checks if the string ends with a specified suffix

### Built-in Functions for Lists

#### Value-inserting methods

- `append()`: Adds an element at the end
- `insert()`: Inserts an element at a specified position

#### Value-removing methods

- `remove()`: Removes the first occurrence of a value
- `pop()`: Removes and returns an element by index

#### Search-related methods

- `index()`: Returns the index of the first occurrence of a value
- `count()`: Returns the number of occurrences

#### Arrangement-related methods

- `sort()`: Sorts the list in ascending order
- `reverse()`: Reverses the order of elements

## Errors

Errors in programming are unexpected issues that disrupt the proper execution of a program.

### Syntax Errors

Mistakes in the structure or "grammar" of the code.

```
if x = 5: # Incorrect use of '=' instead of '=='  
    print("x is 5")
```

### Runtime Errors

Errors that occur during execution (e.g., dividing by zero)

```
a = 10  
b = 0  
result = a / b # Division by zero
```

### Logical Errors

The code runs without crashing but produces incorrect results.

```
side_length = 5  
# Incorrect formula for calculating the area of a square (uses  $\pi$   
r^2)  
area = 3.14 * (side_length ** 2)  
print("Area of the square:", area)
```

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